A Hybrid Heuristic Algorithm for Optimizing the Optimal Landing Time Window Problem in Intelligent Air Transportation Systems

Yong Lu, Shikang Chen, and Xiuqin Pan

***Abstract*— The Aircraft Landing Problem (ALP) involves optimizing the scheduling of flight arrivals and departures while simultaneously managing airport resources to maximize flight utilization, minimize delays, reduce costs, and enhance overall operational efficiency. However, existing algorithms for solving ALP often face challenges in converging optimally and effectively handling penalty values associated with timing violations. To address these issues, this paper proposes a novel hybrid heuristic algorithm, AGWOA, designed to enhance convergence rates and effectively manage penalties in ALP. AGWOA integrates Artificial Bee Colony (ABC) and Genetic Algorithm (GA) techniques into the Whale Optimization Algorithm (WOA), leveraging their complementary strengths to strengthen global search efficiency and refine local constraint handling. This integration accelerates convergence and significantly mitigates penalty costs. Moreover, AGWOA incorporates an adaptive coefficient to facilitate improved convergence, along with a Fast Convergence Update Mechanism (FCUM) to guide the algorithm toward optimal solutions more efficiently. Experimental results conducted on public datasets demonstrate that AGWOA outperforms existing advanced algorithms in both convergence speed and penalty minimization. Specifically, AGWOA achieves a 4.05% decrease in penalty costs compared to baseline algorithms. These results underscore AGWOA's effectiveness in overcoming the challenge of slow convergence and its competitive advantage over other methods in optimizing ALP. The proposed algorithm offers a promising solution for real-world ALP applications, significantly enhancing airline operational efficiency and optimizing resource management.**

***Index Terms*— ALP, WOA, heuristic algorithm, problem optimization, intelligent air transportation systems.**

I. Introduction

THE rapid and sustained growth in global air transportation has positioned airports as essential hubs in contemporary transportation systems. However, the soaring costs associated with expanding airport infrastructure demand innovative approaches to maximize the efficiency of existing resources. This scenario introduces the Aircraft Landing Problem (ALP)—a complex, multi-objective scheduling challenge central to modern aviation management.

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ALP requires the precise coordination of aircraft landings through the simultaneous assignment of runways and scheduling of landing times under a host of rigorous constraints. [1], [2] [3], [4], [5]. In practical terms, airlines must manage increasing flight volumes, comply with stringent landing time constraints, and minimize penalties associated with delays and inefficient resource allocation. Traditional optimization methods, such as linear and dynamic programming, are inadequate in this context as they fail to effectively address the non-linear, multi-dimensional nature of ALP.

In contrast, metaheuristic algorithms have emerged as powerful alternatives in solving these complex challenges. Over the past decades, these algorithms have demonstrated their ability to explore large solution spaces and provide near-optimal solutions in a computationally efficient manner [6], [7], [8]. The Whale Optimization Algorithm (WOA), inspired by the foraging behavior of humpback whales, has gained significant recognition due to its simple structure and effective balance between global exploration and local exploitation [9], [10], [11], [12], [13]. Despite these advantages, existing metaheuristic approaches—including WOA—often encounter challenges such as slow convergence and inefficient handling of penalty constraints when applied to the highly constrained environment of ALP.

The "no free lunch" theory postulates that no single algorithm can universally solve NP-hard optimization problems [14]. Recognizing these limitations, this study introduces a hybrid heuristic algorithm, AGWOA, which seamlessly combines the strengths of the Artificial Bee Colony (ABC) and Genetic Algorithm (GA) into the WOA framework. By enhancing global search capabilities and refining local convergence strategies, AGWOA is designed to more effectively address the intricacies of ALP optimization. Ultimately, this approach aims to achieve a more robust and efficient solution for optimizing aircraft landing schedules, thereby reducing delays, minimizing penalty expenditures, and improving overall airport operations.

This study aims to assess AGWOA's performance in ALP through simulation experiments and numerical tests, evaluating its ability to adapt to complex multi-constraint scenarios. The paper will include a comprehensive analysis of AGWOA's effectiveness in addressing ALP, examining both its strengths and limitations. Through this research, we aim to introduce novel strategies for optimizing ALP, thereby enhancing

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aviation operational efficiency and offering practical decision support for related fields.

The primary contributions of this paper are as follows:

a) Development of a Hybrid Heuristic Algorithm (AGWOA): A novel hybrid heuristic algorithm, AGWOA, is introduced, integrating the strengths of WOA, ABC, and GA. This approach enhances the global search capability and local convergence efficiency of WOA, effectively addressing its inherent limitations, such as slow convergence and suboptimal penalty management, particularly in complex, multi-constraint optimization problems like ALP.

b) Application of AGWOA to ALP: The proposed AGWOA is applied to ALP, demonstrating superior performance compared to existing algorithms in the field. Extensive experimental evaluations reveal that AGWOA achieves improved convergence rates and reduced penalty values, significantly optimizing aircraft landing scheduling by enhancing runway utilization and minimizing delays.

c) Enhancement of Adaptive Coefficient: The adaptive coefficient in WOA is further refined within AGWOA, enabling a more effective balance between exploration and exploitation during the optimization process. This enhancement accelerates convergence and decreases penalty values, ultimately improving the overall solution quality for ALP.

d) Introduction of a Fast Convergence Update Mechanism (FCUM): A key feature of AGWOA is the FCUM, designed to expedite the algorithm's convergence rate. FCUM efficiently guides offspring whales toward optimal positions, significantly reducing computational time while ensuring the attainment of high-quality solutions within a shorter time frame.

The introduction in Section I briefly outlines AGWOA and ALP. Section II lists related work in recent years. Section III provides a theoretical analysis of AGWOA. Section IV details the logic and framework of AGWOA. Section V presents and analyzes the results. Section VI provides conclusions and future directions.

II. Literature Review

In various studies on the Aircraft Landing Problem (ALP), significant efforts have focused on addressing the complex issues of landing scheduling and taxiway planning, with multiple optimization algorithms and models developed to address challenges related to uncertainty, conflict management, and resource utilization optimization. In 2021, Katsigiannis et al. investigated the allocation of airport slots in the context of extreme airport congestion by considering multi-objective and multi-level factors [15]. The study introduced a novel Tri-Objective Slot Allocation Model (TOSAM), which considers total scheduling displacement, maximum scheduling displacement, and demand-based fairness. To solve this problem, a multi-level, multi-objective algorithm was proposed. In 2024, Pang et al. introduced an innovative machine learning-based method for optimizing aircraft landing schedules under conditions of uncertainty [16]. Using flight trajectory and event data from the Sherlock database of the Atlanta Air Route Traffic Control Center (ARTCC ZTL), the effectiveness of this method was demonstrated in practical applications. It was shown that the proposed method reduced total landing time by an average of 17.2% compared to the First-Come, First-Served (FCFS) rule.

Moreover, although the application of metaheuristic algorithms to solve the Aircraft Landing Problem (ALP) has been relatively uncommon in recent years, notable progress has been achieved. In 2020, Zheng et al. introduced a Hybrid Simulated Annealing with Reduced Variable Neighborhood Search (HSARVNS) algorithm aimed at finding near-optimal solutions [17]. Through extensive benchmarking of randomly generated Aircraft Scheduling and Parking Problem (ASPP) instances and a simplified version of ASPP, the HSARVNS algorithm was demonstrated to find optimal solutions for small-scale instances in a minimal amount of time and high-quality solutions for instances involving up to 250 aircraft within a reasonable time frame, thereby demonstrating its effectiveness and efficiency. That same year, Androutsopoulos et al. presented a hybrid heuristic algorithm that combined the Feasibility Pump (FP) algorithm with Large Neighborhood Search (LNS) techniques [18]. By generating new problem instances from real slot request datasets of Greek Regional Airports (GRA), the performance of the algorithm was assessed. The results indicated that the algorithm demonstrated reasonable accuracy and effectively approximated the entire feasible boundary of the problem. Additionally, Salehipour introduced a straightforward heuristic algorithm that decomposed the problem into a series of smaller [1], more manageable subproblems and considered dynamic factors such as the continuous variation in the number of arriving flights and the short time window for determining landing schedules. It was demonstrated in the study that this algorithm was capable of achieving satisfactory solutions within a short time, meeting the demand for rapid response.

Messaoud conducted a systematic review of the Aircraft Landing Problem (ALP) in 2021 [19], demonstrating how the application of metaheuristic methods could assist operators in improving their landing queue management decisions, thereby reducing solution costs. Consequently, numerous advanced metaheuristic algorithms have been proposed to find high-quality solutions for ALP, and the pooling principle of several metaheuristic or exact methods suggests that it could foster the development of more successful heuristic approaches. In the same year, Pohl et al. introduced an optimization model for runway scheduling during winter operational conditions [20], which included the development of pruning rules and effective inequalities, and employed heuristic methods to derive initial solutions. This approach enabled the computation of runway schedules that significantly reduced aircraft delay costs within seconds. Additionally, Huo et al. addressed the challenges of aircraft scheduling under uncertainty by applying a simulated annealing algorithm combined with a time-decomposed sliding window and conducting a case study on Paris Charles de Gaulle Airport [21]. The research offered a robust scheduling solution for arriving flights.

In 2022, Pohl et al. re-examined the challenge of runway scheduling in winter conditions [22], with particular attention to scenarios requiring temporary runway closures for snow, ice, and slush removal. They employed a constraint programming-based initial heuristic to generate feasible initial

TABLE I
SUMMARY OF RELATED WORK

Year	Author(s)	Contribution	Approach/Algorithm	Key Findings	Limitations
2020	Zheng et al.	Hybrid simulated annealing and reduced variable neighborhood search for aircraft scheduling	HSARVNS	Found near-optimal solutions for small-scale instances and high-quality solutions for up to 250 aircraft.	Struggles with scalability for larger datasets; may not be efficient in highly dynamic scenarios.
2020	Androutsopoulos et al.	Hybrid heuristic combining Feasibility Pump with Large Neighborhood Search for slot request datasets	Feasibility Pump and Large Neighborhood Search	Improved accuracy and approximated the feasible boundary effectively.	Limited to specific slot request datasets; performance drops for larger, more complex problems.
2020	Salehipour	Heuristic approach to dynamic and smaller subproblems for aircraft scheduling	Heuristic Algorithm	Achieved satisfactory solutions in a short time, suitable for rapid responses in dynamic conditions.	Limited by its application to smaller subproblems; may not address large-scale, multi-objective ALP scenarios effectively.
2021	Messaoud	Systematic review of ALP and metaheuristic methods	Metaheuristics	Demonstrated how metaheuristic methods reduce solution costs in ALP landing queue management.	General review without focusing on specific algorithm improvements; lacks practical validation of suggested approaches.
2021	Katsigiannis et al.	Allocation of airport slots in extreme congestion conditions	Tri-Objective Slot Allocation Model (TOSAM)	Proposed a multi-level, multi-objective algorithm to improve slot allocation in congested airports.	Does not consider real-time dynamic scheduling changes; computational complexity increases with higher congestion levels.
2021	Pohl et al.	Runway scheduling during winter operations	Pruning rules and inequalities	Reduced aircraft delay costs significantly within seconds using heuristic methods.	Not adaptable to non-winter operational conditions; may struggle with highly variable weather scenarios.
2021	Huo et al.	Aircraft scheduling under uncertainty with simulated annealing	Simulated Annealing	Provided robust solutions for airport scheduling at Paris Charles de Gaulle Airport.	Limited applicability outside of the specific airport context; may require more robust handling of extreme uncertainty.
2022	Pohl et al.	Winter runway scheduling with temporary runway closures	Constraint programming and column generation	Outperformed other methods for winter runway scheduling under closure conditions.	Limited to winter conditions with runway closures; performance may degrade under different operational scenarios.
2022	Deng et al.	Taxiway planning using multi-strategy hybrid optimization	MPSACO	Introduced a pheromone hybrid approach to enhance optimization performance in airport taxiway planning.	Focuses mainly on taxiway planning, limiting its application to broader scheduling problems; performance may degrade with more complex constraints.
2024	Pang et al.	Optimizing landing schedules under uncertainty	Machine learning-based method	Reduced total landing time by 17.2% compared to FCFS, using data from ARTCC ZTL.	May not handle extreme outliers or unexpected events effectively; reliance on machine learning models may require substantial data preprocessing.

solutions and utilized a column generation approach to identify all variables necessary for solving the binary program. This approach outperformed time-continuous formulations and pure constraint programming methods. In the same year, Deng et al. introduced a Multi-Strategy Particle Swarm and Ant Colony Hybrid Optimization Algorithm (MPSACO) to address the problem of airport taxiway planning [23]. They developed a novel pheromone initialization method employing the Cooperative Wolf-Based Particle Swarm Optimization (CWBPSO) algorithm, devised an innovative pheromone allocation mechanism, and developed a pheromone update strategy based on the wolf pack hunting principle. These strategies were integrated into a new pheromone hybrid approach, enhancing the influence of optimal solutions' pheromones, dynamically adjusting the search direction, and avoiding a decrease in optimal search capability, thereby significantly improving the algorithm's optimization performance. In studies concerning ALP, metaheuristic algorithms, when compared to traditional methods, have demonstrated superior performance in both solution speed and quality, showcasing their potential and advantages in handling large, complex problems. These studies not only provide more precise and efficient scheduling solutions for airport management but also contribute to the advancement of ALP and related domains. A summary of related work is shown in Table I.

III. METHODOLOGY

As an emerging heuristic optimization algorithm, WOA has demonstrated promising applications in addressing a wide

range of practical challenges. Researchers have extensively investigated various optimization algorithms aimed at enhancing the efficiency and precision of flight scheduling for the Aircraft Landing Problem (ALP). This section introduces relevant studies on WOA, the Artificial Bee Colony (ABC) algorithm, Genetic Algorithm (GA), ALP, and aircraft landing algorithms, providing a summary of advancements in optimizing aircraft landing operations.

The rationale behind selecting WOA lies in its robust global search capabilities. WOA navigates the search space adeptly, circumventing local optima by emulating the bubble-net attacking strategy of whales during hunting. However, WOA encounters challenges in managing complex multi-constraint problems. To address this, we bolster its local search capabilities and diversity by integrating the Artificial Bee Colony (ABC) algorithm and Genetic Algorithm (GA), thereby enhancing its overall performance.

In the hunting process of WOA, humpback whales target the best current position as prey, adjusting individual positions through trend matching and spiral hunting actions. During the encircling phase, whales interact with the optimal agent and update positions using Equation (4), with Equations (1), (2), and (3) governing parameter calculations. The adjustment of positions towards optimal solutions relative to the current position is facilitated by vectors A and C. These mechanisms are formalized as follows [24]:

a) Mathematical model for encircling prey:

$$D = |C \cdot X_{\text{best}} - X_i(t)| \tag{1}$$

$$A = 2 \cdot a \cdot r_1 - a \tag{2}$$

$$C = 2 \cdot r_2 \tag{3}$$

$$X_i(t+1) = X_{\text{Best}} - A \cdot D \tag{4}$$

In this context, X_{Best} denotes the best position discovered thus far. The parameter a linearly decreases from 2 to 0 across iterations, facilitating the contraction of the encircling behavior during both exploration and exploitation phases. Here, r_1 and r_2 represent random numbers within the interval [0, 1].

b) Mathematical model for the spiral bubble-net predation:

The bubble-net attacking strategy primarily incorporates two mechanisms: the shrinking encircling mechanism and the logarithmic spiral position updating. The shrinking encircling mechanism updates the humpback whale's position according to Equation (4) [25], while Equation (5) modifies the calculation method for parameter D by reducing the value of a in Equation (6). The whale progressively tightens the encircling circle while following the spiral path described in Equation (7). The position update formulas are as follows:

$$D' = |X_{\text{Best}} - X_i(t)| \tag{5}$$

$$X_i(t+1) = D' \cdot e^{bl} \cdot \cos(2\pi l) + X_{\text{Best}} \tag{6}$$

D' represents the distance between the whale and its prey (the best solution obtained so far), where l is a random number within the range [1, 1], and b is a constant that defines the logarithmic spiral shape. Assuming a 50% probability of selecting the spiral model during the shrinking process, Equation (7) is derived. The mathematical model for the logarithmic spiral position update is as follows:

$$X_i(t+1) = \begin{cases} X_{\text{Best}} - A \cdot D & \text{if } p < 0.5, \\ D' \cdot e^{bl} \cdot \cos(2\pi l) + X_{\text{Best}} & \text{if } p \geq 0.5. \end{cases} \tag{7}$$

where p is a random number in the range [0, 1].

Furthermore, to explore the global optimum, the algorithm selects individuals randomly to guide position updates, utilizing the change in vector A for searches. In contrast to the exploitation phase, where updates rely on the current best search agent, Equations (8) and (9) stipulate the random selection of search agents. This behavior is defined as follows [26]:

$$D = |C \cdot X_{\text{rand}}(t) - X_i(t)| \tag{8}$$

$$X_i(t+1) = X_{\text{rand}}(t) - A \cdot D \tag{9}$$

where X_{rand} is the position of a randomly selected whale in the current whale population.

Recent research on WOA includes several notable advancements. In 2020, Guo et al. introduced an enhanced WOA incorporating sociological learning and a wavelet mutation strategy [27]. Their approach introduced a novel linear increment probability to enhance global search capabilities. Also in 2020, Got et al. proposed the Guided Population Archive Whale Optimization Algorithm (GPAWOA) [28], which utilizes an external archive to store non-dominated solutions, thereby enhancing optimization outcomes. In 2021, Chakraborty et al. presented the Enhanced WOA (WOAmM) [29], integrating WOA with a modified Symbiotic Organisms Search (SOS) to address premature convergence issues and achieve a more balanced algorithm. Building on this work, in 2022, Chakraborty et al. introduced the Improved WOA (ImWOA) [30], which focuses on enhancing solution diversity to mitigate local optima and maintain solution accuracy. Also in 2022, Yang et al. developed the MAWOA-SSELM model [31], leveraging Multi-Agent Whale Optimization Algorithm (MAWOA) to optimize the selection of parameters for the Spectral-Spatial Enhanced Logistic Model (SSLEM), demonstrating promising accuracy in well logging layer recognition. In 2023, Tian et al. proposed a multi-objective disassembly line balancing optimization model based on a discrete WOA, [32] with the objective of minimizing idle time, improving smoothness, reducing cost, and lowering energy consumption during disassembly operations. The model integrates K-means clustering and local search strategies to enhance the algorithm's accuracy and was effectively implemented to address the disassembly problem of a worm reducer.

Next, ABC stands out as a widely adopted optimization algorithm successfully applied across various practical problems. It has found application in numerical, binary, integer, mixed-integer, and combinatorial optimization contexts [33]. In ABC, bees search for optimal nectar sources, where candidate solutions represent multiple food sources tailored to the problem at hand. Through iterative processes, bees strive to discover the optimal solution.

In the employed bee search phase, employed bees encircle each solution $X_i (i = 1, 2, \ldots, \text{SN})$ and endeavor to discover superior solutions [34]. This stage, represented by

Equation (10), constitutes a pivotal aspect of the algorithm, wherein diligent efforts of employed bees persistently explore the solution space and enhance current solutions. This iterative refinement and optimization process are crucial factors that facilitate ABC's gradual convergence towards the global optimum, guiding the algorithm towards achieving optimal solutions.

$$V^*_{i,jr} = x_{i.jr} + \emptyset_{i,jr} \cdot (x_{i,jr} - x_{k,jr}) \tag{10}$$

where X_k is randomly selected from the entire population $X_k \neq X_i$, jr is a random integer within the range $[1, D]$. The weight $\varnothing_{i,jr}$ is randomly generated between -1 and 1 to ensure diversity and flexibility in the search. Following the process described by Equation (10), each employed bee generates a new solution, potentially representing an improved solution discovered during the search process.

$$V_{i,j} = \begin{cases} V^*_{i,jr}, & \text{if } j = jr, \\ x_{i,j}, & \text{otherwise.} \end{cases} \tag{11}$$

Here, $j = 1, 2, \ldots, D$. Subsequently, a greedy selection method is utilized to acquire an improved solution. This approach ensures that at each iteration, the direction most advantageous to optimizing the objective is chosen, thereby driving the algorithm steadily towards enhanced solutions.

$$X_i = \begin{cases} V_i, & \text{if } V_i \text{ outperforms } X_i, \\ X_i, & \text{otherwise.} \end{cases} \tag{12}$$

In the scout bee phase that ensues, each new solution is compared with its parent solution, and the superior one is chosen. Scout bees remain passive until they share the quality of solutions within the hive. The activation of scout bees is contingent upon a trial counter, with its maximum threshold fixed and tested against limits—a distinctive control parameter of ABC. New food sources generated by Equation (10) are then assigned to scout bees, and the counter is reset.

GA, on the other hand, is regarded as one of the most significant optimization algorithms, inspired by Darwin's theory of natural selection. GA's primary operators include selection, crossover, and mutation [35]. At the outset of this process, fitness functions such as root mean square error (RMSE) serve as metrics to search for a generation of elite populations. Fitness evaluation functions determine the fitness value of each individual, with superior values carried over to the next generation while weaker ones are discarded. This stage can be represented as [36]:

$$G_i(p\{x_0, \ldots, x_b\}) = \begin{bmatrix} p_0 \\ \vdots \\ p_n \end{bmatrix} = \begin{bmatrix} x_{00} & \cdots & x_{0b} \\ \vdots & \ddots & \vdots \\ x_{n0} & \cdots & x_{nb} \end{bmatrix} \tag{13}$$

Subsequently, crossover facilitates the exchange of genetic information between parents, which is then passed on to the next generation. Mutation further enhances diversity in subsequent generations. The general framework diagram of the above method is shown in Figure 1.

In WOA, its primary appeal lies in its simple yet powerful search mechanism, capable of swiftly identifying optimal solutions [37]. During the spiral search phase in WOA, whales explore the solution space by spiraling towards the optimal solution. This spiral search path resembles the circling approach aircraft use during the landing phase, where they circle around the target landing point. Both processes involve deliberate exploration of the search space to pinpoint the optimal landing point or solution. In WOA's spiral phase, whales progressively reduce the radius of the spiral path to approach the optimal solution. Similarly, aircraft during the circling landing phase gradually adjust their flight path to approach the landing point.This iterative approximation process enables the identification of optimal solutions within a constrained search space. Additionally, the spiral search mechanism in WOA demonstrates adaptability and flexibility, permitting whales to modify their search strategies in response to the problem's characteristics and objectives. Similarly, during the aircraft circling landing phase, pilots must adapt their flight paths in response to environmental conditions and the location of the target landing point. This adaptability and flexibility imbue both WOA and the aircraft circling landing phase with robust adaptive capabilities.

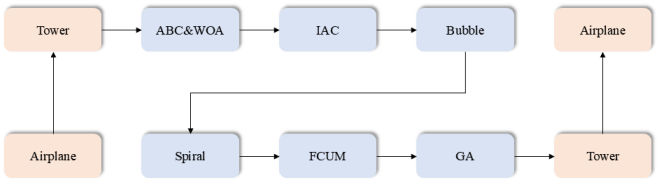


Fig. 1. Method framework diagram.

The rationale for selecting WOA stems from its robust global search capability, well-suited for addressing complex optimization problems. However, WOA faces challenges in managing multiple constraints and local optimization. To address these limitations, we incorporate ABC and GA into WOA to bolster its capabilities in local search and diversity. Furthermore, enhancements such as improved adaptive coefficients and the Fast Convergence Update Mechanism (FCUM) are introduced to expedite algorithmic convergence and enhance adaptability to dynamic changes.

ABC simulates the foraging behavior of bees to explore the solution space, where bees investigate potential solutions and select regions with higher fitness based on pheromone concentration. This approach, inspired by the foraging behavior of bees in honey collection, proves effective for various mixed optimization problems [38]. Similarly, ALP requires searching for optimal landing times within the solution space to meet constraints and optimization goals. ABC achieves diverse search and local optimization by utilizing roles such as employed bees, onlookers, and scouts. In ALP, this parallels roles like employed whales, observer whales, and scout whales, enhancing search diversity and facilitating improved solutions through local optimization. Fitness evaluation in ABC measures the distance and quality between bee dances and food sources, while in ALP, it assesses the quality of each landing time solution and the satisfaction of constraints. Both processes necessitate defining fitness functions tailored to specific problems and objectives, guiding the search based on these evaluations.

Integrating GA enhances the algorithm's robustness, enabling it to effectively navigate various problem types and complex search spaces. Compared to other outstanding multi-objective algorithms, GA achieves solutions with superior diversity and convergence [39]. This multi-algorithm approach effectively mitigates local optima and enhances the algorithm's global search capability, thereby accelerating convergence.

AGWOA combines the strengths of WOA, ABC, and GA to optimize ALP effectively. WOA excels in global search, identifying optimal solutions across the entire search space. ABC complements this by performing robust local searches within specific regions, while GA enhances diversity and exploration capabilities through selection, crossover, and mutation operations. AGWOA integrates WOA's rapid convergence and adaptability with ABC's diverse search and local optimization, tailored to ALP's adaptive requirements. This multi-algorithm integration balances global and local search spaces, mitigates the risk of local optima, and enhances overall algorithm performance and efficiency. Additionally, incorporating GA enhances the algorithm's robustness and adaptability to handle diverse problem types and complex search spaces. By leveraging the comprehensive strengths of these three algorithms, AGWOA achieves improved convergence to final solutions with minimized penalty values, significantly enhancing algorithmic performance and efficiency.

At the same time, an optimization strategy for ALP was introduced. Similar to the foraging process in ABC, the aircraft's search for an appropriate landing time during the landing process is considered. When searching for an appropriate landing time, aircraft must consider factors such as constraints and optimization objectives, meaning that they must search within the possible solution space to find a suitable landing time that satisfies these constraints and objectives. Analogous to ABC, different roles for whales, such as employed whales, onlooker whales, and scout whales, can be introduced to perform corresponding tasks, increasing search diversity and finding better solutions in local searches. These whales can adjust parameters such as their path or speed to achieve the optimal landing time, thereby meeting specific constraints and optimization objectives. The fitness evaluation reflects the quality of each landing time scheme and the degree to which it satisfies the constraints. Similar to the fitness evaluation in ABC, the fitness evaluation in ALP is defined according to the specific problem and objectives, guiding the search process based on the evaluation results. The organic combination of WOA, ABC, and GA can create a more comprehensive and efficient optimization framework to tackle various types of problems and complex search spaces, thereby enhancing the performance and efficiency of the algorithm. The overall algorithm framework is illustrated in Figure 2.

IV. AGWOA

AGWOA integrates WOA, ABC, and GA to strike a balance between global search and local optimization. Sections *A* and *B* will provide detailed descriptions of the components and specific implementation of AGWOA.

A. Improved Adaptive Coefficient

Early convergence is a prevalent challenge in optimization algorithms, particularly when tackling complex or high-dimensional problems. In WOA, the linear decrease of coefficient a to 0 can impede the algorithm's convergence rate and constrain both local and global optimization capabilities. Furthermore, linear relationships may inadequately capture the search for optimal solutions amidst multiple constraints. Introducing a nonlinearly decreasing inertia weight enables the algorithm to avoid premature convergence to local optima, thereby enhancing opportunities to discover the global optimum. Therefore, a is refined to:

$$a = \left(\frac{t}{T_{\max}}\right)^{\beta} \tag{14}$$

Choose T_{max} and t, combined with the exponent β, to ensure adaptability of the algorithm. The algorithm adjusts the inertia weight dynamically based on the current iteration number t and the maximum iteration number T_{max}, thereby adapting its search strategy as the algorithm progresses. Here, t represents the current iteration number, directly reflecting the algorithm's advancement. Using t as the numerator ensures that the inertia weight changes in accordance with the algorithm's progress, thereby dynamically adjusting the search strategy. T_{max} serves as the denominator, setting clear boundaries and defining the range of the inertia weight throughout the entire search process. The exponent β, as the gradient parameter, enables nonlinear adjustments that capture pivotal points in the search process, facilitating adaptive search strategy adjustments. By adjusting the value of β, the algorithm controls the rate of inertia weight reduction, ensuring nonlinear variation and enhancing flexibility and adaptability.

Equation (14) addresses the limitation of linear inertia weight decrease in traditional WOA, aiming to enhance the algorithm's global search capability and convergence speed. The linear reduction of inertia weight in standard WOA iterations can lead to excessive focus on local optima in later stages, limiting exploration of the global solution space. Nonlinear adjustment provides a more balanced approach to exploration and exploitation throughout the search process. It facilitates rapid reduction of the weight during early iterations to bolster global search, while gradually decreasing it in subsequent iterations to refine solutions, aligning closely with practical algorithm needs.

B. Integration of Fast Convergence Update Mechanism (FCUM)

After applying genetic operations from GA to update whale positions, the offspring whales no longer follow their parents' paths, introducing randomness and increasing the risk of falling into local optima. To address this issue, the Fast Convergence Update Mechanism (FCUM) is introduced. In FCUM, the parent generation moves along a convergent path. Subsequently, the offspring update their positions by choosing directions with larger convergence gradients rather than moving aimlessly. The core principle behind FCUM is to guide offspring towards the direction of the currently identified optimal solution post-genetic operations. This approach

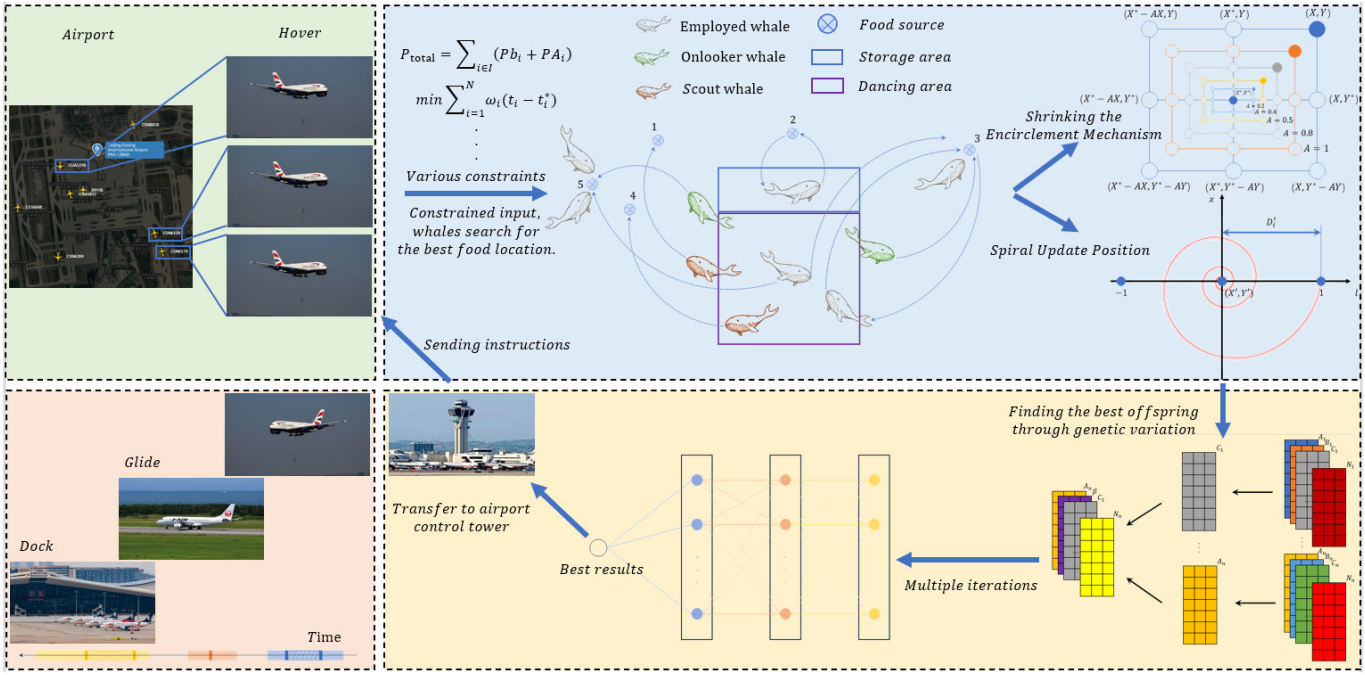


Fig. 2. Algorithm principle.

accelerates convergence towards the vicinity of the optimal solution while mitigating randomness in the search process. If an updated position proves worse than the current one, a smaller step size is chosen to stabilize the search and prevent falling into local optima.

Let $X_i(t)$ be the position of the offspring t at time i, and $X_i^*(t+1)$ be the position of individual t after one update from $X_i(t)$. If there exists a number $\eta \in [0, 1)$, such that:

$$|X_i^*(t+1) - X_i(t)| \le \eta |X_i^*(t) - X_i(t-1)| \tag{15}$$

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$$\eta \ge \frac{|X_i^*(t+1) - X_i(t)|}{|X_i^*(t) - X_i(t-1)|} \tag{16}$$

This mechanism updates the offspring's position towards $X_i^*(t+1)$. Otherwise, it takes the position corresponding to the minimum value of η. This mechanism accelerates convergence speed by iterating the convergence of the parent generation and optimizing the offspring generation. As depicted in Figure 3, the blue lines illustrate the convergence path under the FCUM mechanism, while the gray lines represent the original convergence path.

Proof is as follows:

For $\{X_i(t)\}$, we have

$$\begin{aligned}
&|X_i^*(t+p) - X_i(t)| \\
&\le |X_i^*(t+p) - X_i(t+p-1)| + \cdots + |X_i^*(t) - X_i(t-1)| \\
&\le (\eta^{n+p-1} + \cdots + \eta^n)|X_i^*(1) - X_i(0)| \\
&\le \frac{\eta^{n+p-1}\left(1 - \eta^{1-p}\right)}{1-\eta}|X_i^*(1) - X_i(0)| \\
&\le \frac{\eta^n}{1-\eta}|X_i^*(1) - X_i(0)|
\end{aligned} \tag{17}$$

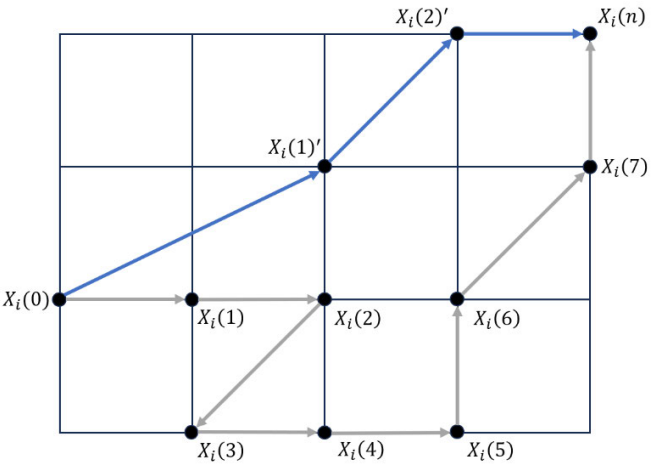


Fig. 3. Schematic representation of FCUM.

That is, $\lim_{n\to+\infty} \frac{\eta^n}{1-\eta}|X_i^*(1) - X_i(0)| = 0 \iff \forall \epsilon > 0, \exists N$, when $n > N$, there is:

$$|X_i^*(t+p) - X_i(t)| \le \frac{\eta^n}{1-\eta}|X_i^*(1) - X_i(0)| < \epsilon \tag{18}$$

C. The Improved Basic Process

In AGWOA, the population is initially initialized, and then the fitness value function, using Equation (1) and the greedy method to select better food sources, is computed where $f(X_i)$ denotes the fitness value of each solution:

$$X_{\text{best}} = \text{argmax}\, f(X_i) \tag{19}$$

The probability $P_{\text{follow}}(X_i)$ that each solution X_i is found by the follower bees is given by:

$$P_{\text{follow}}(X_i) = \frac{f(X_i)}{\sum_{j=1}^{N} f(X_j)} \tag{20}$$

The probability $P_{\text{choose}}(X_i)$ that each solution X_i is selected as a reference point is given by:

$$P_{\text{choose}}(X_i) = \frac{f(X_i)}{\max_{j=1}^{N} f(X_j)} \tag{21}$$

where $\max_{j=1}^{N} f(X_j)$ represents the highest fitness value among all solutions.

Thus, in AGWOA, the probability that prey is chosen by the following whales is represented as the product of Equation (20) and Equation (21):

$$P_{\text{total}}(X_i) = \frac{f^2(X_i)}{\sum_{j=1}^{N} f(X_j) \times \max_{j=1}^{N} f(X_j)} \tag{22}$$

The subsequent Equation (4) will be updated to:

$$X_{\text{encircle}}(t+1) = X_{\text{nearest_whale}} - (A+B) \times C \tag{23}$$

where $X_{\text{nearest_whale}}$ represents the position of the nearest whale to the prey, A and C denote parameters from WOA and ABC respectively, and B signifies the influence factor of the bubble-net. The update of the logarithmic spiral position is represented by Equation (7), updated to:

$$X_{\text{new}}(t+1) = \begin{cases} X_{\text{best}} - (A+B) \times D & \text{if } p < 0.5, \\ D' \times e^{bl} \times \cos(2\pi l) + X_{\text{best}} & \text{if } p \geq 0.5. \end{cases} \tag{24}$$

The subsequent GA crossover and mutation operations are used to generate new individuals, performed after repeating Equation (19).

$$X_{\text{child}} = \text{Crossover}(X_{\text{parent}_1}, X_{\text{parent}_2}) \tag{25}$$

$$X_{\text{mutant}} = \text{Mutation}(X_{\text{child}}) \tag{26}$$

where X_{child} represents the generated offspring individual, $Crossover$ denotes the crossover operation function that performs crossover on two parent individuals X_{parent_1} and X_{parent_2} to produce X_{child} as offspring. X_{mutant} is the individual undergoing mutation, and $Mutation$ is the operation function that mutates X_{child} to obtain X_{mutant}. Equation (10) updates the position of the scouting whale. The flowchart and pseudo code are shown in Section D.

Each solution (whale individual) represents a sequence of aircraft landing times encoded as a vector $[t_1, t_2, \ldots, t_N]$. Initially, a set of random solutions is generated to satisfy basic constraints. The fitness value of each solution is computed based on the objective function, with a primary focus on total delay time and penalties. The WOA's bubble net attack strategy is employed for global exploration in the search space to discover new solutions. ABC and GA are introduced for local search to optimize solutions within specific regions. ABC conducts detailed searches in local areas through selection, crossover, and mutation operations to enhance solution diversity. Finally, the search strategy dynamically adjusts based on the current solution quality to accelerate the convergence process. These steps are repeated until termination conditions

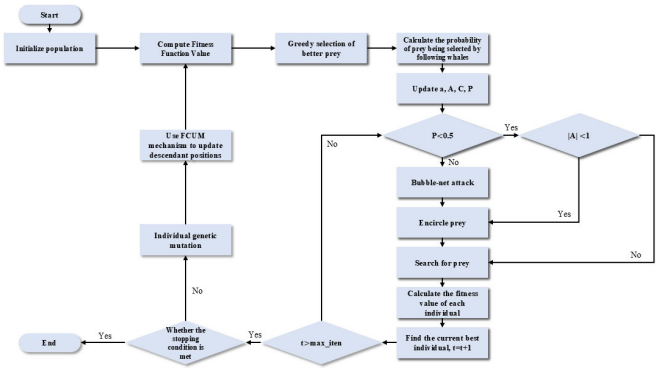


Fig. 4. Flowchart of AGWOA.

are met (such as maximum iteration count or convergence criteria). Through these steps, AGWOA effectively optimizes ALP, finding high-quality solutions that satisfy multiple constraints.

AGWOA's time complexity primarily consists of several components: the generation of the initial population, which is $O(N)$ where N is the population size; the time complexity of fitness evaluation, $O(N \cdot M)$, where M is the dimensionality of each solution (number of aircraft); global search complexity, $O(N \cdot T)$, where T is the maximum number of iterations; and local search complexity, $O(N \cdot k)$, where k represents the number of operations in local search. Therefore, the overall time complexity can be expressed as $O(N \cdot M + N \cdot T + N \cdot k)$.

AGWOA offers significant advantages over other advanced heuristic and metaheuristic algorithms. It achieves a balanced approach between global search and local optimization by integrating WOA, ABC, and GA, resulting in notable improvements in convergence speed and solution quality. Moreover, AGWOA excels in managing dynamic changes and uncertainties, effectively addressing diverse challenges in practical applications through FCUM and adaptive adjustment capabilities.

D. Basic Process

The flowchart is illustrated in Figure 4, and the pseudocode is shown in Algorithm 1. Initially, several parameters are set, including the maximum iteration count, adaptive coefficient parameters (a, A, c), and the probability parameter (p), which controls the trade-off between exploration and exploitation. Additionally, mutation and crossover rates, which regulate the operations of the genetic algorithm, are initialized. The population of whales is initialized randomly, along with the mutation and crossover operators that will be utilized during the optimization process.

The main loop of the algorithm begins by selecting the optimal food sources from the population, representing the most promising solutions. For each whale in the selected group, a probability is computed to determine whether an exploration or exploitation move will be executed. If the probability value (p) is less than 0.5, the whale performs an exploitation search in the current space by updating its position using either a local or global search strategy. Specifically, if parameter A, which influences the movement's direction and

magnitude, is less than 1, the whale applies a local update rule based on (21). Otherwise, a global search update rule, as defined in (9), is employed. If p is greater than or equal to 0.5, the whale executes a global search by updating its position using (22).

Once the whale updates its position, the new coordinates are verified to confirm that they remain within the feasible search space while adhering to problem constraints. The algorithm then identifies the optimal whale in the current population, representing the best solution identified thus far. This iterative process continues for a specified number of cycles or until the predefined convergence criterion is satisfied.

Following each iteration of the main loop, the population undergoes mutation and crossover operations. Mutation introduces small random changes to the whale's position, thereby maintaining population diversity and mitigating premature convergence. Crossover, in contrast, combines characteristics from two whales to generate new offspring that inherit the most advantageous traits from both. These operators facilitate a more thorough exploration of the search space and enhance the algorithm's convergence toward an optimal solution. Once the mutation and crossover operations are completed, the fittest offspring is incorporated into the next generation.

This iterative process—updating whale positions, verifying feasibility, applying mutation and crossover, and identifying the best offspring—continues for a predetermined number of iterations. The algorithm proceeds until a termination condition is met.

V. Experiment

The ALP is a complex optimization challenge that involves scheduling aircraft arrivals and landings at airports while ensuring that each flight adheres to stringent timing constraints, minimizing delay-related penalties, and optimizing resource utilization. As air traffic continues to grow, the significance of ALP increases, making efficient scheduling crucial for alleviating congestion and improving airport operational efficiency. In this study, we address the optimization of aircraft landing schedules with the objective of balancing flight scheduling and penalty minimization within stringent operational constraints.

The experiment applies AGWOA to model ALP using datasets from OR-Library. It employs various parameter settings, including iteration numbers, population sizes, and configurations, to comprehensively assess and analyze AGWOA's performance. The study aims to deepen understanding of the algorithm's convergence, stability, and exploration capabilities across solution spaces by comparing different parameter variations.

To evaluate the effectiveness of the proposed AGWOA, we compare it with several well-established baseline algorithms, each selected for its relevance and efficiency in solving the ALP. These algorithms encompass a range of metaheuristic and hybrid optimization approaches, providing a comprehensive benchmark for assessing performance. The following baseline algorithms are examined in this study:

Salehipour's Algorithm (2020) [1]: Already introduced in Part 2.

Algorithm 1 AGWOA Pseudocode

Initialization:
Initialize $max_{iteration}$, a, A, C, p, $mutation_{rate}$, $crossover_{rate}$
X_{best} = optimal search position
initialize population P
initialize mutation operator M
initialize crossover operator X
Main:
$iteration = 0$
while (Not meet the stopping criterion) **do**
  $selected_{food}$ = select better food(P)
  for each whale in $selected_{food}$ **do**
    calculate probability(whale, P)
    $t = 0$
    while ($t < max_{iteration}$) **do**
      update parameters(whale, a, A, C, p)
      if ($p < 0.5$) **then**
        if ($|A| < 1$) **then**
          update position using Eq.(21)
        else
          update position using Eq.(9)
        end if
      else
        update position using Eq.(22)
      end if
      ensure valid position(whale)
      find optimal individual(P)
      $t = t + 1$
      if ($t > max_{iteration}$) **then**
        break
      end if
    end while
  end for
  if (Not meet the stopping criterion) **then**
    for each whale in P **do**
      perform mutation(whale, M, $mutation_{rate}$)
      perform crossover(whale, X, $crossover_{rate}$)
    end for
    update best offspring(P)
  end if
  $iteration = iteration + 1$
end while

Hybrid Metaheuristic by Hassan and Pillay [40]: This hybrid metaheuristic approach combines multiple search strategies, designed to enhance solution quality in complex ALP scenarios. By integrating local and global search mechanisms, this algorithm achieves superior solution quality compared to traditional metaheuristics. Comparing AGWOA with this algorithm highlights its ability to balance exploration and exploitation while minimizing computational costs.

ISA Algorithm by Hammouri et al. [41]: The Iterated Simulated Annealing (ISA) algorithm integrates Iterated Local Search (ILS) with Simulated Annealing (SA) to address the multiple-runway ALP. It efficiently navigates the solution space to minimize penalties and optimize runway scheduling.

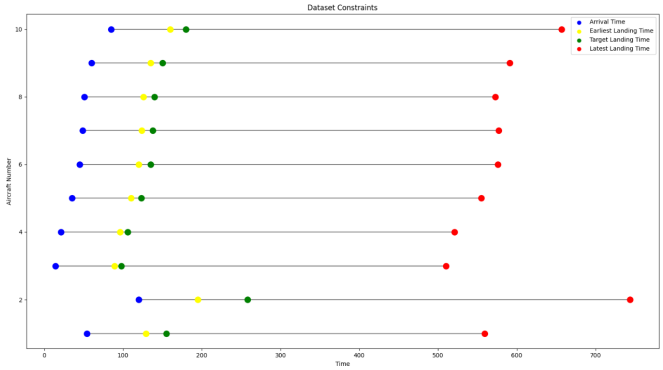


Fig. 5. Illustration of time constraints.

ISA's strong local search capability establishes it as a strong benchmark for assessing AGWOA's local search performance, particularly regarding penalty minimization and convergence speed.

Multi-Objective Optimization by Zhang et al. [42]: This approach employs multi-objective optimization to manage the conflicting objectives of the ALP, such as minimizing delay times, optimizing runway utilization, and reducing fuel consumption. The method incorporates criteria selection and multi-objective optimization to achieve a balanced trade-off among competing factors. Comparing AGWOA with this algorithm facilitates the evaluation of its capability to address complex, multi-objective problems while ensuring convergence and reducing penalties.

This experiment aims to explore the potential of WOA in solving ALP, offering more effective scheduling solutions for the aviation transportation sector. Additionally, the study investigates and applies optimization algorithms to address practical and complex problems.

A. Dataset

For the experimental evaluation, datasets from the OR-Library, a well-known repository providing test cases for various optimization and operational research problems, including ALP, are utilized. The dataset utilized in this study includes essential data points necessary for solving the aircraft landing scheduling problem. These data files are formatted with aircraft count (p) and freeze time. For each plane i ($i = 1, \ldots, p$): arrival time, earliest landing time, target landing time, latest landing time, penalty per unit landing time, and penalty cost per landing unit time. Following these, for each plane j ($j = 1, \ldots, p$): required separation time.

B. Experiment Process

In the dataset description, each aircraft in the nth dataset group has $n + 6$ constraints. The first six constraints are fixed and describe the aircraft's landing timeline and penalties, while the subsequent n constraints describe the landing time intervals for n aircraft. The time constraints are illustrated in Figure 5.

AGWOA incorporates a dynamic adjustment mechanism designed to handle uncertainties such as weather changes and unexpected delays. With the introduction of real-time updated FCUM, AGWOA promptly adjusts the positions and search directions of whales based on current environmental data. Furthermore, AGWOA adapts algorithm parameters dynamically to accommodate evolving constraints and objectives, thereby ensuring flexibility and robustness in aircraft landing scheduling.

TABLE II
SYMBOLS OF RELATED CONSTRAINTS

Constraint Symbol	Constraint Name
i	Aircraft
I	Aircraft set
A_i	Arrival time
E_i	Earliest landing time
T_i	Target landing time
L_i	Latest landing time
TE_i	Time difference before target landing
TL_i	Time difference after target landing
Pb_i	Penalty cost per unit time before target
PA_i	Penalty cost per unit time after target
P_{total}	Total penalty cost
S_{ij}	Separation time between aircraft i and j
x_i	Landing time of aircraft
x_i^*	Expected landing time of aircraft
$\Delta_{\min}$	Minimum interval time
$d_{i,j}$	Distance between aircraft i and j
D_{safe}	Safe distance

The primary objectives of ALP are to minimize total delay time, reduce fuel consumption, and optimize runway utilization at airports. Key constraints include minimum aircraft separation times, requirements for takeoff and landing time differences, and penalty costs. AGWOA integrates these objectives and constraints into its fitness function, utilizing robust global and local search capabilities to achieve an optimal balance in the multi-dimensional search space. Specifically, AGWOA effectively manages multi-constraint conditions through enhanced adaptive coefficients and the fast convergence update mechanism, ensuring optimal scheduling. A detailed breakdown of these constraints is provided in Table II below.

Given the time constraints defined by its earliest and latest landing times, the optimal landing time for a specific aircraft i is assumed to be its earliest landing time to minimize penalty costs associated with delays:

$$P_{\text{total}} = \sum_{i \in I} (Pb_i + PA_i) \tag{27}$$

The remaining constraints can be expressed as [43]:
Minimize total delay time:

$$\min \sum_{i=1}^{N} \omega_i (t_i - t_i^*) \tag{28}$$

Minimum separation time between aircraft:

$$x_{i+1} - x_i \geq \Delta_{\min} \tag{29}$$

Priority scheduling of aircraft:

$$t_i \geq t_i^* \tag{30}$$

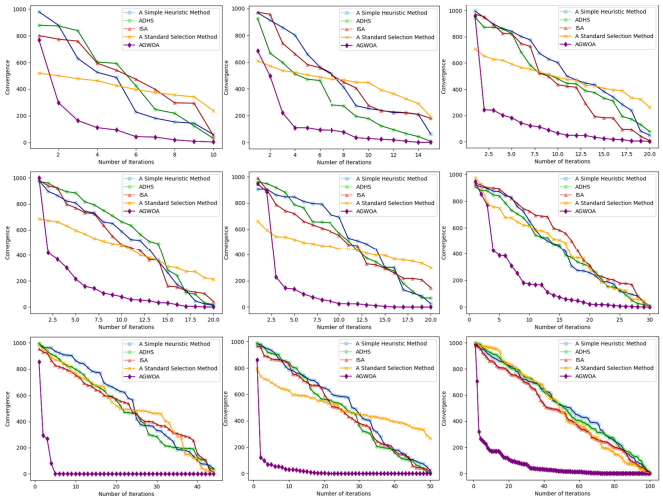


Fig. 6. Iteration count and convergence plot.

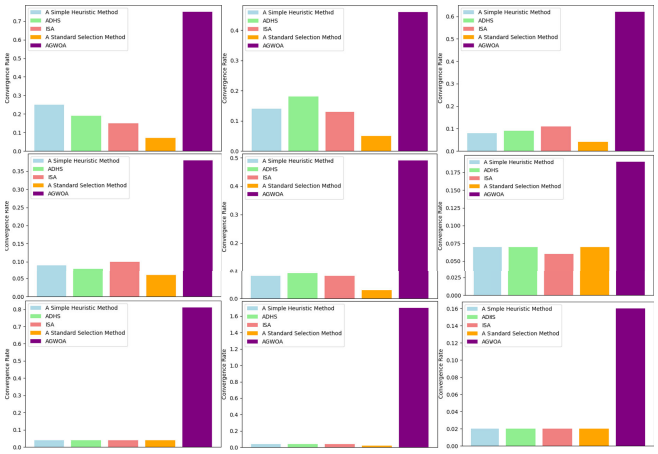


Fig. 7. Average convergence rate.

Safety distance requirement:

$$d_{i,j} \geq D_{\text{safe}} \tag{31}$$

And:

$$E_i \leq x_i \leq L_i, \quad i \in I \tag{32}$$

$$0 \leq \text{TE}_i \leq T_i - E_i, \quad i \in I \tag{33}$$

$$0 \leq \text{TL}_i \leq L_i - T_i, \quad i \in I \tag{34}$$

$$\text{TE}_i \geq T_i - x_i, \quad i \in I \tag{35}$$

$$\text{TL}_i \geq x_i - T_i, \quad i \in I \tag{36}$$

$$x_i = T_i - \text{TE}_i - \text{TL}_i, \quad i \in I \tag{37}$$

C. Experimental Results

The iteration count and convergence behavior of the algorithm are illustrated in Figure 6. The nine images depict the convergence behavior of the algorithm's iterations across various aircraft numbers, corresponding to a simple heuristic method [1], the ADHS [40], the ISA [41], a standard selection method [42], and the AGWOA algorithm. The images are sequentially arranged from left to right and top to bottom. It is apparent that AGWOA exhibits markedly superior convergence compared to other advanced algorithms in the same category.

Based on the convergence analysis of the aforementioned algorithms, the average convergence rate across different datasets can be additionally observed, as shown in Figure 7. It is evident that AGWOA demonstrates superior convergence performance across various datasets. Such convergence behavior is crucial in optimization problems, as it signifies that the algorithm can find high-quality solutions within a reasonable time frame, thereby affirming its potential and efficacy in practical applications.

In small-scale datasets, the penalty value λ resulting from Amir Salehipour's simple heuristic method [1] is used as the baseline. The relative penalty Δ for other algorithms is calculated accordingly:

$$\Delta(\%) = \frac{\lambda' - \lambda}{\lambda} \times 100\% \tag{38}$$

TABLE III
COMPARISON OF RESULTS ON SMALL-SCALE DATASETS

Dataset	A Simple Heuristic Method [1]	ISA [41]	AGWOA
Airland1	0	0	0
Airland2	0	0	-3.37
Airland3	0	0	-14.48
Airland4	0	0	-9.52
Airland5	0	0	-9.51
Airland6	0	0	-9.90
Airland7	0	0	-10.12
Airland8	0	0	-7.02

TABLE IV
COMPARISON OF RESULTS FOR LARGE-SCALE DATASETS

Algorithm Name	OR-Library	MASP-SET	FPT
A Simple Heuristic Method [1]	0	0	0
ADHS [40]	3.70	0.74	0.72
ISA [41]	1.15	4.62	5.05
Standard Selection Method [42]	8.17	0.75	0.72
AGWOA	1.39	-0.47	-0.68

A negative result indicates that the penalty value is lower compared to the reference algorithm. The comparison of penalty values for small-scale instances is presented in Table III below.

AGWOA performs comparably to the benchmark algorithm and ISA in the first dataset, but outperforms both in the remaining datasets from the small-scale instances.

In addition to the large-scale datasets provided by the OR-Library, this study further evaluates algorithm performance using two additional large-scale datasets, MASP-SET and FPT. As shown in Table IV, the simple heuristic method proposed by Amir Salehipour [1] serves as the baseline, and penalty values for alternative algorithms are computed based on Equation (38).

In large airport scenarios, AGWOA has demonstrated superior computational performance. Across extensive instances, AGWOA achieves penalty values closely following the benchmark algorithm and ISA, with a difference of only 0.24% compared to ISA, showcasing high stability and consistency across multiple experiments. Additionally, AGWOA exhibits

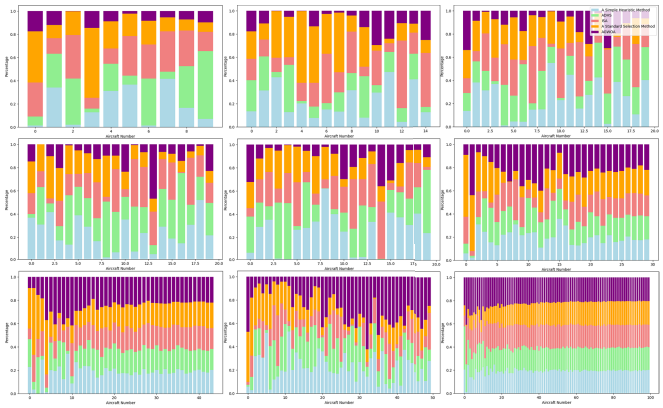


Fig. 8. The difference between the algorithm's results and the target time.

excellent scalability in handling larger-scale problems through optimized algorithm structures and parallel processing, effectively managing increased flight scheduling tasks. Analysis of experimental results indicates that AGWOA excels in solving ALP, significantly enhancing optimal solutions by reducing aircraft scheduling costs, maximizing resource utilization, and optimizing flight arrangements.

By comparing the computed results to each aircraft's target landing time and assessing the differences, the accuracy of the proposed algorithm can be evaluated. This difference provides a direct measure of the algorithm's capacity to meet practical requirements in solving the ALP. Figure 8 shows the distribution of these differences, which demonstrates that AGWOA displays smaller time deviations in the context of the ALP. This suggests that AGWOA offers a notable advantage in aligning with the target landing time, more effectively addressing real-world operational demands, and underscoring its potential for complex airspace management.

To rigorously assess the overall efficiency of the proposed five algorithms, we conducted statistical analysis using the Friedman test. The Friedman test, a non-parametric method, is well-suited for comparing median differences among multiple related samples, making it particularly useful for evaluating algorithm performance. Each algorithm was executed on multiple aircraft landing instances, with total penalty values recorded for each instance and algorithm combination. These penalty values were structured into a matrix format where rows represent test instances and columns represent algorithms. The Friedman test was employed to analyze performance discrepancies among algorithms across various instances.

Specifically, assuming a significance level of $\alpha = 0.05$, the Friedman test revealed significant differences among algorithms with a Friedman statistic χ_F^2 resulting in a p-value well below the threshold. The approximate p-value of 0 allowed us to reject the null hypothesis, signifying notable performance distinctions among the algorithms. Subsequent post-hoc analysis affirmed AGWOA's superior performance across most instances, followed by the ISA algorithm, as detailed in Table V below.

This analysis not only underscores the efficacy of AGWOA but also elucidates the performance differentials among algorithms in addressing ALP, thereby offering a scientific

TABLE V
AVERAGE RANKINGS OF COMPARISON METHODS CALCULATED BY FRIEDMAN TEST

Algorithm Name	Ranking
AGWOA	2.22
A Simple Heuristic Method [1]	2.33
ISA [41]	2.33
ADHS [40]	3.67
A Standard Selection Method [42]	4.44

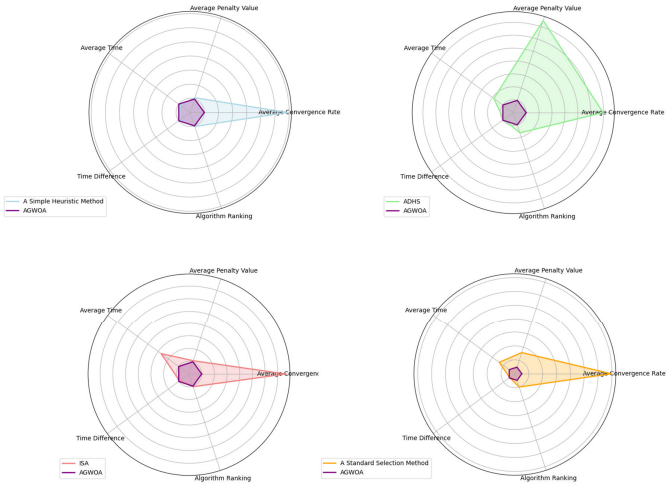


Fig. 9. Radar chart of multidimensional comparison.

foundation for selecting the optimal algorithm. The outcomes of the Friedman test and subsequent post-hoc analysis distinctly outline the comparative strengths of various algorithms in enhancing punctuality, optimizing runway utilization, mitigating air traffic controller workload, and ensuring equity among airlines.

In practical applications, runtime is a critical metric for assessing algorithm performance. We conducted a thorough comparison of runtime between AGWOA and other advanced algorithms. AGWOA demonstrates efficient computational performance, especially in managing larger-scale problems through optimized algorithm structures and parallel processing. It consistently delivers high-quality solutions within stringent real-world time constraints. Our experiments were conducted on a standard PC featuring an Intel(R) Core(TM) i5-8250U CPU @ 1.60 GHz with 8GB of RAM, running the Windows 11 operating system, with runtime measurements recorded in seconds, as detailed in Table VI.

A comprehensive comparison of all key dimensions demonstrates that AGWOA exceeds the performance of other comparable algorithms. Figure 9 summarizes these comparison results, depicting AGWOA's performance with respect to convergence speed, penalty value, runtime, and deviations from the target landing time. As depicted in the figure, AGWOA not only demonstrates notable advantages in optimization speed and accuracy but also excels in minimizing computational resource consumption and enhancing overall efficiency. This indicates its considerable adaptability and robustness for addressing complex optimization challenges.

TABLE VI
COMPARISON OF ALGORITHM EXECUTION TIMES

Dataset Name	A Simple Heuristic Method [1]	ADHS [40]	ISA [41]	A Standard Selection Method [42]	ADS-B [44]	Benders [45]	AGWOA
Airland1	0.003	0.007	0.005	0.006	0.005	0.007	0.006
Airland2	0.005	0.008	0.001	0.002	0.005	0.005	0.001
Airland3	0.007	0.008	0.005	0.004	0.011	0.007	0.008
Airland4	0.006	0.003	0.006	0.004	0.010	0.006	0.001
Airland5	0.006	0.008	0.005	0.001	0.010	0.009	0.002
Airland6	0.009	0.002	0.002	0.005	0.009	0.005	0.009
Airland7	0.021	0.019	0.015	0.015	0.020	0.024	0.019
Airland8	0.259	0.217	0.137	0.358	0.362	0.220	0.167
Airland9	11.174	24.457	35.887	36.531	21.920	28.672	13.051

The experimental results of this study conclusively demonstrate that AGWOA shows considerable promise in effectively solving ALP, thereby offering a robust optimization tool for the aviation transportation sector. However, to address more complex real-world applications, further research and exploration are essential to enhance the algorithm's efficiency and broaden its applicability.

VI. CONCLUSION AND OUTLOOK

Significant potential has been demonstrated by AGWOA in optimizing the ALP, showcasing robust search capabilities and adaptability in managing complex constraints. The risk of local optima is mitigated through a balance of global exploration and local optimization, offering competitive solutions for efficient flight scheduling. Aviation operational efficiency is enhanced through its ability to address various scheduling challenges.

Although promising results have been observed with small-scale datasets, further enhancements are necessary for its application to large-scale problems. Future research should be directed toward refining local search strategies and enhancing the algorithm's adaptability to handle more complex ALP instances. An evaluation of its performance under diverse constraint conditions would further broaden its applicability.

The implementation of AGWOA in real-world flight scheduling systems should be further investigated. Its effectiveness in practical airline operations should be assessed to gain valuable insights into its real-world utility and impact, ultimately contributing to more efficient solutions for the aviation industry.

The advancement of ALP optimization through AGWOA holds great promise, and continued research is expected to refine its capabilities and provide valuable contributions to the aviation sector.

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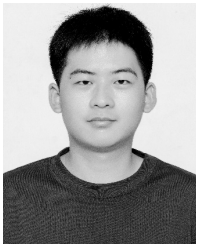
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